

Can Qualifying Position Predict F1 Race Results?



- Research Question: Can Formula 1 qualifying position predict race finishing position?
- Qualifying sets the starting grid, but race-day factors (strategy, weather, incidents) still matter.
- Main Analysis: Formula 1 data from 2023–2025.
- **Conclusion: Qualifying position is a strong predictor, but not a perfect one.**

Data Sources

Source	Type	Role	Coverage	Notes
OpenF1 API	REST API	Main source	2023–2025	Best recent coverage
FastF1	Python library	Validation source	2023–2025	Historical session data
Kaggle F1 Dataset	CSV dataset	Validation source	Partial recent	Some recent data incomplete

Key variables: qualifying_position, race_position, driver, year, circuit

All sources standardized into one merged analysis schema for fair comparison.

Main Result: Strong Positive Relationship

03

1,208

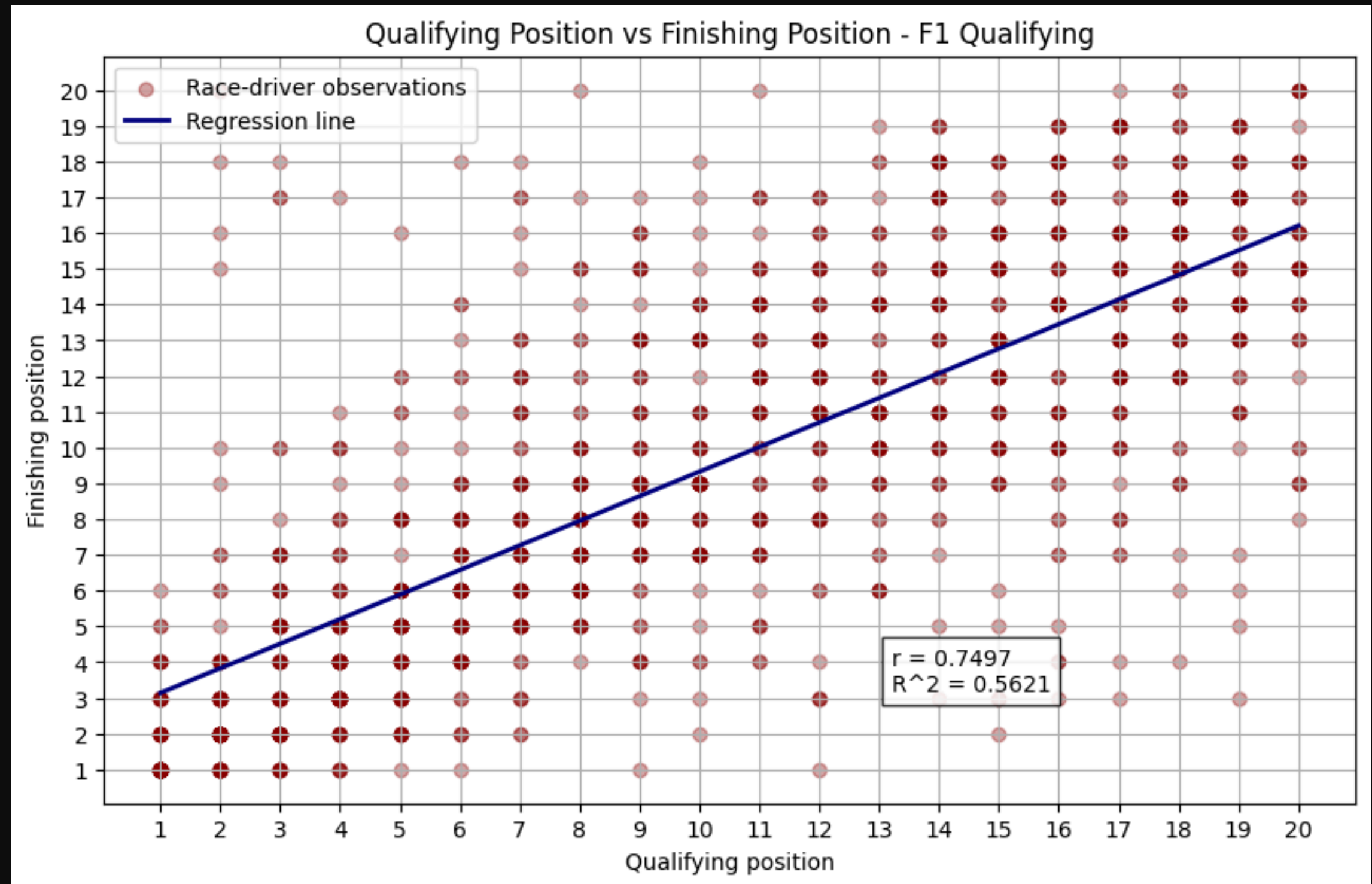
Driver-Race
Observations

69

Races
Analyzed

0.7497

Pearson
Correlation



Scatter: Qualifying Position vs. Race Finishing Position

Regression & Predictive Strength

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Model: $\text{race_position} = 0.6877 \times \text{qualifying_position} + 2.4485$

R²

0.5621

Qualifying explains 56% of race outcome variation

MAE

2.6405

Mean Absolute Error (positions)

RMSE

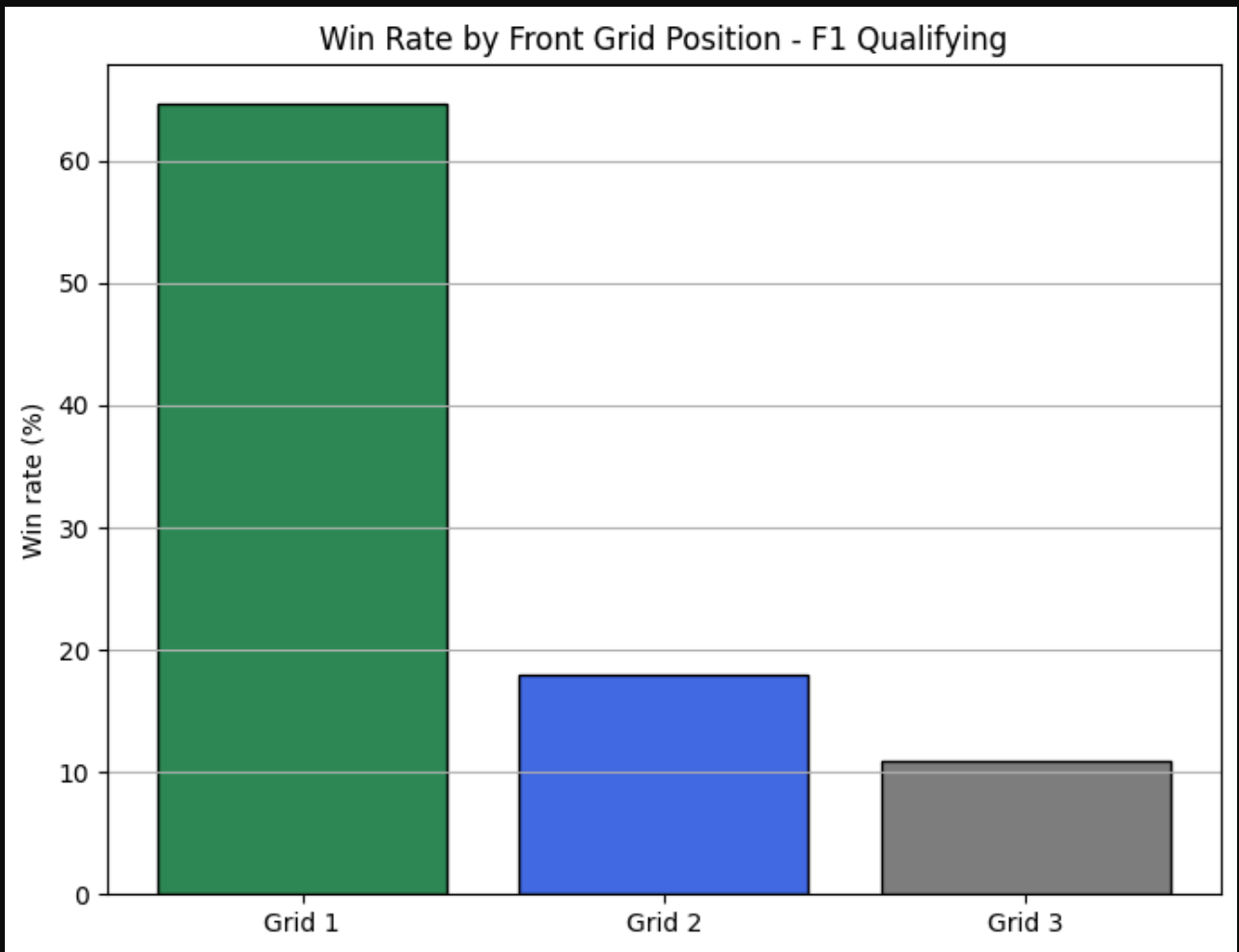
3.4669

Root Mean Squared Error (positions)

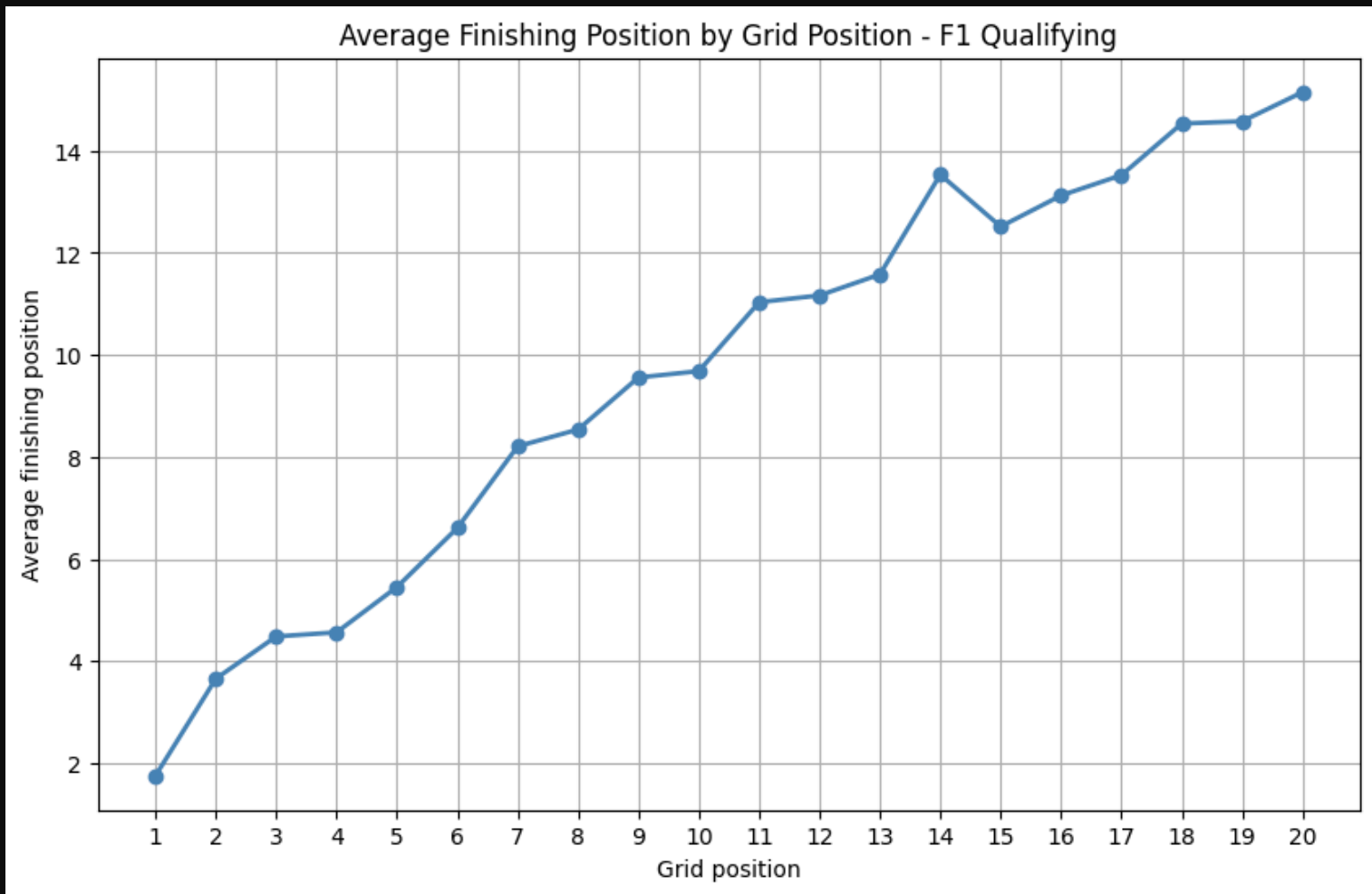
Key Insight: Qualifying position explains a meaningful portion of race outcomes ($R^2 = 0.5621$), but race-day factors — strategy, reliability, and incidents — still account for the rest.

Pole Position Advantage

64.6% Pole-to-Win Rate 42 wins out of 65 races



F1_Pole_to_Win_barchart



F1_Qualifying_linechart

Cross-Source Comparison

Data Source	Records	Races	Correlation (r)
OpenF1	1,208	69	0.7497
FastF1	958	48	0.6649
Kaggle	299	15	0.6751



All three sources consistently show a positive relationship — confirming the main conclusion.

Challenges & Conclusion

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Challenges

- Different data sources used different formats and field names
- Historical coverage was inconsistent, especially for recent Kaggle qualifying/results data
- The hardest part was data cleaning and standardization across sources — not the regression itself

Conclusion

Qualifying position is a strong predictor of Formula 1 race finishing position.

This pattern is consistent across multiple data sources, although qualifying alone cannot fully explain outcomes.

Strategy · Reliability · Incidents · Weather — race-day factors still play important roles.